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# Hydrodynamics of helical bacteria with polar flagella

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## ABSTRACT

Amphitrichous bacteria possessing a helical cell body propel themselves by coordinating the rotation of two helical flagella anchored at opposite poles of a spirillum-shaped body. The motility modes of these cells depend on the relative rotational directions of the two polar flagella and the magnitude of the motor torque. Here, we develop a mathematical model of an amphitrichous bacterium to elucidate its swimming mechanisms in a viscous fluid. The hydrodynamics of the system is governed by the incompressible Stokes equations, subject to instantaneous force- and torque-free constraints, with flagellar motor rotation providing the sole actuation. For a helical bacterium with a single polar flagellum, we identify four dynamically stable swimming modes—pull, push, wrapping, and overwhirling—consistent with those observed for rod-shaped cells. We further show that swimming efficiency is greater when the cell body and flagellum have opposite helical handedness than when they share the same handedness. Extending the model to two oppositely anchored flagella, we systematically characterize the swimming states arising from different combinations of motor rotation and torque magnitude and delineate the coordination regimes that yield stable propulsion. Finally, we incorporate external magnetic fields through an applied magnetic torque and demonstrate that magnetic and hydrodynamic torques compete to control cell orientation, with stronger fields increasingly aligning swimming trajectories with the applied field.

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## I. INTRODUCTION

Bacteria swim in viscous, low-Reynolds number environments in which inertial effects are negligible and propulsion arises from the nonreciprocal motion of slender appendages.[1–3] The primary mode of bacterial locomotion is the rotation of one or more flagella, slender helical filaments driven by reversible rotary motors embedded in the cell envelope.[2,4–7] In this mechanism, swimming dynamics emerges from the interplay between flagellar rotation and the counterrotation of the cell body, mediated by viscous hydrodynamic interactions with the surrounding fluid. Experimental and theoretical studies have further elucidated how variations in motor rotation and flagellar configuration give rise to forward and backward runs, reorientation, and complex trajectory changes.[2,8–16]

Bacteria that swim by flagellar rotation share common structural features of the flagellum. Each flagellum consists of a long, slender helical filament connected to the cell body via a short hook.[3,5,17–19] The equilibrium configuration of the filament can be characterized by its curvature and twist and is well described by elastic rod models.[20–22] The hook, which is significantly more flexible than the filament, acts as a mechanical coupling that transmits motor torque while allowing bending and torsional compliance during swimming.[23,24] However, species-specific differences in flagellar structure and spatial localization lead to distinct swimming behaviors. For example, *Pseudomonas putida* possesses intrinsically left-handed polar flagella that generate push, pull, and wrapping modes, whereas *Escherichia coli* exhibits multiple polymorphic helical states distributed over the cell body that facilitate transitions between run and tumble phases.[25,26]

Beyond flagellar actuation, cell body geometry plays an important role in locomotion by modifying hydrodynamic resistance and swimming direction.[15] Bacteria exhibit diverse cell morphologies. For example, *Escherichia coli*[1,3] and *Bacillus subtilis* possess rod-shaped


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cell bodies,[27] whereas curved-rod morphologies are observed in species such as *Vibrio cholerae*.[28] Other organisms adopt helical,[6,29] spirochete,[30] or teardrop-shaped geometries.[31] Amphitrichous bacteria, exemplified by *Magnetospirillum magneticum* AMB-1, possess a helical body with two polar flagella.[29] In such swimmers, coordination of the two flagella generates propulsion: for example, directed translation occurs when one flagellum pushes and the other pulls, whereas reversal of both motors induces swimming reversal.[18] The hydrodynamic interplay between helical body geometry and dual polar actuation yields characteristic swimming patterns.[18,32]

In addition, AMB-1 aligns and swims along geomagnetic field lines. This behavior arises from intracellular chains of magnetosomes that endow the cell with a permanent magnetic dipole moment.[29,33] An externally applied magnetic field therefore exerts a torque on the cell body, biasing its orientation while leaving the propulsion mechanism controlled by flagellar coordination. This separation of propulsion and orientation enables magnetic steering without altering the underlying propulsion mechanism. Magnetotactic bacteria thus provide a natural system in which internally generated hydrodynamic torques couple with externally imposed magnetic torques, offering a tractable framework for examining how morphology and external fields jointly regulate locomotion in viscous flows.[29,34]

Rod-shaped and spherical cell body geometries with helical flagella have served as canonical model systems in theoretical and experimental studies of chemotaxis and run–tumble dynamics.[5,13,14,16,25,35,36] However, computational investigations of the hydrodynamics of bacteria with a helical cell body and helical flagella remain largely unexplored. In this work, we develop a mathematical model based on the geometry of magnetotactic bacterium AMB-1 and employ the regularized Stokeslet formulation to investigate the swimming dynamics of a helical cell with either a single polar flagellum or dual polar flagella. Swimming modes are classified according to motor rotation direction and torque magnitudes, and geometric conditions that optimize propulsion speed are identified. We further explore how helical body handedness and externally applied magnetic fields influence propulsion, reorientation, and trajectory patterns, thereby providing a comprehensive framework for understanding the coupled hydrodynamics and magnetic control of bacterial locomotion.

## II. MATHEMATICAL MODEL

We present a mathematical model of a biflagellate bacterium to study its hydrodynamic interactions caused by flagellar rotations. The cell body is modeled as a helical tube with one turn. A single flagellum extends from each pole of the cell body. Each flagellum is typically composed of three parts: a rotary motor, a compliant short hook, and a long helical filament. Figure 1 shows a schematic diagram of our computational model. Here, we consider the cell body as a rigid structure and the flagella as elastic Kirchhoff rods. The cell motion is coupled to a viscous fluid governed by the regularized Stokes equations. Flagellar motors generate torque that drives the flagellar rotation in either the counterclockwise (CCW) or clockwise (CW) direction. CCW rotation of the flagella can be defined using the right-hand rule, with the thumb aligned with the outward normal vector to the cell body surface along the flagellar axis and the curled fingers indicating the direction of rotation. See the schematic diagram of the mathematical model of the bacterium. For a comprehensive mathematical formulation and the numerical scheme, the readers are referred to Lee *et al.*[14] and Olson *et al.*[37]

### A. Dynamics of the rigid cell body

The cell body of a bacterium takes the form of a spiral tube shape with approximately one turn, with both ends enclosed by hemispheres. Since the cell body is modeled as a rigid structure, its motion is characterized by two degrees of freedom: translation and rotation. Consequently, all points on the body exhibit identical translational and rotational motion, maintaining the constraints of rigid body dynamics. To enforce this rigidity within the numerical framework, we adopt a penalty-based approach. We introduce two representations of the cell surface: one that undergoes rigid body motion (translation and rotation) from a reference configuration, and the other that evolves according to the dynamic fluid–structure interaction (FSI) equations. These two configurations are coupled via stiff linear

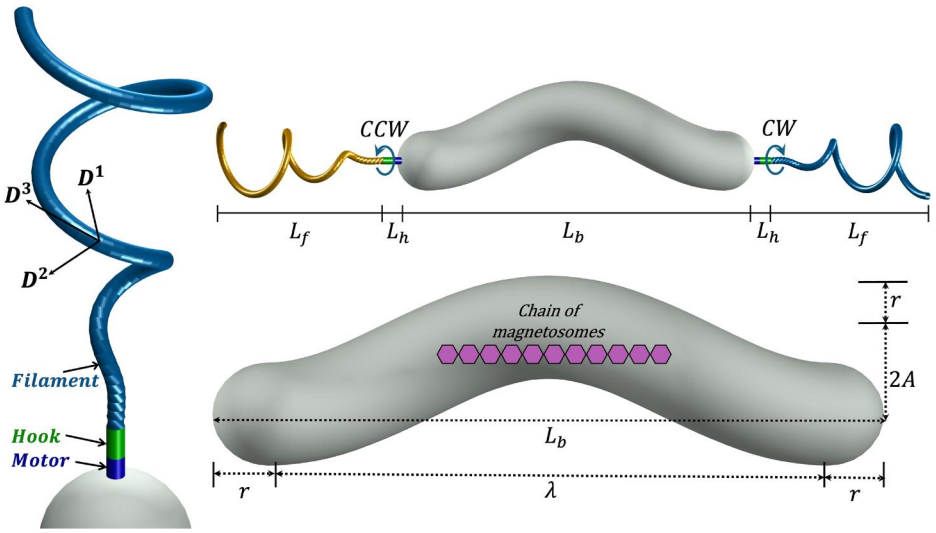



**FIG. 1.** Schematic of a computational model. The cell body is modeled as a left-handed helical tube with both ends capped by hemispherical surfaces and contains an internal chain of magnetosomes. Each helical flagellum comprises a long filament, a short compliant hook, and a rotary motor. The labels CCW and CW denote the rotational directions of flagella imposed by the motor torque at the flagellar base.

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springs, which penalize deviations between them and thereby enforce the rigidity constraints without imposing it exactly.

Let $\mathbf{Z}_i^{\mathrm{B}}, i = 1, \ldots, n_{\mathrm{B}}$, be the reference coordinates of the points that discretize the cell body surface, satisfying $\sum_i^{n_{\mathrm{B}}} \mathbf{Z}_i^{\mathrm{B}} = \mathbf{0}$. The corresponding physical coordinates, $\mathbf{X}_i^{\mathrm{B}}, i = 1, \ldots, n_{\mathrm{B}}$, represent the positions of the points that interact with the surrounding fluid. With a force $\mathbf{f}^{\mathrm{B}}$ and a torque $\mathbf{n}^{\mathrm{B}}$ applied to the rigid body externally, the equilibrium equations are given by

$$\mathbf{F}_i^{\mathrm{B}}(t) = K(\mathbf{X}_i^{\mathrm{B}}(t) - \hat{\mathbf{X}}_i^{\mathrm{B}}(t)), \tag{1}$$

$$0 = \mathbf{f}^{\mathrm{B}}(t) + \sum_{i=1}^{n_{\mathrm{B}}} \mathbf{F}_i^{\mathrm{B}}(t), \tag{2}$$

$$0 = \mathbf{n}^{\mathrm{B}}(t) + \sum_{i=1}^{n_{\mathrm{B}}} (\mathcal{R}(t)\mathbf{Z}_i^{\mathrm{B}}) \times \mathbf{F}_i^{\mathrm{B}}(t), \tag{3}$$

where $\mathbf{Z}_i^{\mathrm{B}} \mapsto \hat{\mathbf{X}}_i^{\mathrm{B}}(t) = \mathbf{T}(t) + \mathcal{R}(t)\mathbf{Z}_i^{\mathrm{B}}$ is the rigid body motion that takes the body from the reference configuration to its current configuration, with translation $\mathbf{T}(t)$ and rotation matrix $\mathcal{R}(t)$, and $K$ is a penalty parameter that acts to enforce the rigidity constraints. Notice that the same translation and rotation are imposed on all of the rigid body points. In the dynamic situation, the configuration $\mathbf{X}^{\mathrm{B}}$ moves with the fluid and is not strictly constrained to move as a rigid body, whereas, by construction, the configuration $\hat{\mathbf{X}}^{\mathrm{B}}$ moves as a rigid body. Penalty forces $\mathbf{F}^{\mathrm{B}}$ act to prevent displacements between these two configurations, and in the limit $K \to \infty$, the configuration $\mathbf{X}^{\mathrm{B}}$ moves as an exactly rigid body.[38]

### B. Unconstrained Kirchhoff rod model for bacterial flagella

To describe the dynamics of flagella, we use an unconstrained Kirchhoff rod model. Each flagellum is modeled as a three-dimensional space curve $\mathbf{X}_n(s,t), n = 1, \ldots, n_{\mathrm{F}}$, accompanied by an orthonormal triads $\{\mathbf{D}_n^1(s,t), \mathbf{D}_n^2(s,t), \mathbf{D}_n^3(s,t)\}$, where $s$ is a Lagrangian parameter along the flagellum, $t$ represents time, and $n_{\mathrm{F}}$ is the number of flagella. For convenience of explanation, we omit the flagellum index and describe a single flagellum. The space curve here represents the centerline of the rod and the triad along the rod shows the degree of bending and twisting. The initial configuration of the flagella attached to the cell body is constructed using the Higdon helix that is tapered near its anchoring point at the motor,[39]

$$\mathbf{X}(s) = (r(s)\cos(ks), r(s)\sin(ks), s), \qquad 0 \le s \le L_{\mathrm{h}} + L_{\mathrm{f}}, \tag{4}$$

where $k$ is the wave number, $L_{\mathrm{h}}$ and $L_{\mathrm{f}}$ are the lengths of the hook and the filament, respectively. The short hook is assumed to be intrinsically straight, with $r(s) = 0$ for $0 \le s \le L_{\mathrm{h}}$, and the helical filament takes a spiral form with the radius $r(s) = r_{\mathrm{f}}(1 - e^{-c(s-L_{\mathrm{h}})^2})$ for $L_{\mathrm{h}} \le s \le L_{\mathrm{h}} + L_{\mathrm{f}}$, where $c = 2.0$. The helical radius of the filament gradually increases to $r_{\mathrm{f}}$.

The standard Kirchhoff rod model imposes two constraints on the rod, specifically, $\mathbf{D}^3 = \frac{\partial \mathbf{X}}{\partial s}$ and $|\frac{\partial \mathbf{X}}{\partial s}| = 1$. These constraints are difficult to handle when the flagellum interacts with a viscous fluid, as they lead to complex boundary conditions on the fluid equations. Rather than strictly enforcing these constraints, we introduce an elastic energy functional that approximates them while allowing for slight deviations. The functional is given by[10,40,41]

$$E = \frac{1}{2}\int \left[ \sum_{i=1}^{3} a_i \left( \frac{\partial \mathbf{D}^j}{\partial s} \cdot \mathbf{D}^k - \Omega_i \right)^2 + \sum_{i=1}^{3} b_i \left( \mathbf{D}^i \cdot \frac{\partial \mathbf{X}}{\partial s} - \delta_{3i} \right)^2 \right] ds, \tag{5}$$

where $(i,j,k)$ is any cyclic permutation of $(1,2,3)$, and $\delta_{3i}$ is the Kronecker delta. The first summation represents the bending energies with bending moduli $a_1$ and $a_2$ of the rod about $\mathbf{D}^1$ and $\mathbf{D}^2$, respectively, and the twisting energy with twist modulus $a_3$. The second summation accounts for shearing energies with shear constants $b_1$ and $b_2$, and stretching energy with the stretch constant $b_3$. The standard Kirchhoff rod theory is recovered when $b_i \to \infty$ for $i = 1, 2, 3$. We adopted literature values for the bending and twist moduli $(a_1, a_2, a_3)$ of *E. coli* flagella.[42–45] The remaining elastic parameter values were selected such that the flagellar filament preserves its initial length to within 0.5% and the deviation angle between $\partial\mathbf{X}/\partial s$ and $\mathbf{D}^3$ remains within 1° during forward and backward runs. The intrinsic strain vector $(\Omega_1, \Omega_2, \Omega_3)$ represents the intrinsic curvature $\kappa = \sqrt{\Omega_1^2 + \Omega_2^2}$ and the intrinsic twist $\Omega_3$. These intrinsic properties determine the geometry of the flagellum's helix, including its pitch and radius. They define the flagellum's stress-free configuration, for which the elastic energy is zero.

Let $\mathbf{F}$ and $\mathbf{N}$ be the internal force and moment, respectively, transmitted across a cross section of the flagellum, the balance equations for linear and angular momentum are given by

$$0 = \mathbf{f} + \frac{\partial \mathbf{F}}{\partial s}, \qquad 0 = \mathbf{n} + \frac{\partial \mathbf{N}}{\partial s} + \frac{\partial \mathbf{X}}{\partial s} \times \mathbf{F}, \tag{6}$$

where $\mathbf{f}$ and $\mathbf{n}$ are the force and torque densities, respectively, applied by the fluid to the rod. The internal force and moment on the perpendicular cross section, $\mathbf{F}$ and $\mathbf{N}$, and the applied force and torque densities, $\mathbf{f}$ and $\mathbf{n}$, may be expanded in the basis $\mathbf{D}^1, \mathbf{D}^2, \mathbf{D}^3$,

$$\mathbf{F} = \sum_{i=1}^{3} F_i \mathbf{D}^i, \qquad \mathbf{N} = \sum_{i=1}^{3} N_i \mathbf{D}^i, \tag{7}$$

$$\mathbf{f} = \sum_{i=1}^{3} f_i \mathbf{D}^i, \qquad \mathbf{n} = \sum_{i=1}^{3} n_i \mathbf{D}^i, \tag{8}$$

where a variational argument of the energy functional yields the constitutive relations as follows:

$$N_i = a_i \left( \frac{\partial \mathbf{D}^j}{\partial s} \cdot \mathbf{D}^k - \Omega_i \right), \qquad F_i = b_i \left( \mathbf{D}^i \cdot \frac{\partial \mathbf{X}}{\partial s} - \delta_{3i} \right). \tag{9}$$

The second relations approximately enforce the constraints that the flagellum is inextensible and that $\mathbf{D}^3$ points in the same direction as $\frac{\partial \mathbf{X}}{\partial s}$.

The reference flagellum $\mathbf{X}(s)$ is anchored at each pole of the cell body, positioned normal to the surface at the start and governed by the above balance equations. At each motor point $\mathbf{X}_n(0,0)$, the unit tangent vector aligns with $\mathbf{D}_n^3(0,0)$, while the principal normal and binormal vectors align with $\mathbf{D}_n^1(0,0)$ and $\mathbf{D}_n^2(0,0)$, respectively. Two additional constraints are imposed to ensure that the motor point remains on the cell surface at all times and that the tangent vector $\mathbf{D}_n^3(0,t)$ at $\mathbf{X}_n(0,t)$ stays closely aligned with the normal vector $\mathbf{E}_n(t)$ to the cell surface at the motor point.

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### C. Hydrodynamics of a cell: Regularized Stokes formulation

The bacterium is immersed in an incompressible viscous fluid, and its motion is coupled to the fluid dynamics which is governed by the regularized Stokes equations,

$$0 = -\nabla p + \mu \Delta \mathbf{u} + \mathbf{g}, \qquad 0 = \nabla \cdot \mathbf{u}, \tag{10}$$

where $\mu$ denotes the fluid viscosity. The unknowns $\mathbf{u}$, $p$, and $\mathbf{g}$ correspond to the fluid velocity, pressure, and the external force per unit volume applied to the fluid, respectively. These variables are functions of Cartesian coordinates $\mathbf{x}$ and time $t$. The force density $\mathbf{g}$ exerted on the fluid by the immersed cell is given by

$$\begin{aligned}\mathbf{g}(\mathbf{x},t) = \sum_{n=1}^{n_F}\Bigg[ &\int_0^L [-\mathbf{f}_n(s,t)]\psi_\epsilon(\mathbf{x}-\mathbf{X}_n(s,t))ds \\ &+ \frac{1}{2}\nabla \times \int_0^L [-\mathbf{n}_n(s,t)]\psi_\epsilon(\mathbf{x}-\mathbf{X}_n(s,t))ds \\ &+ \int_0^L [-\mathbf{f}^{\mathrm{r}}_n(s,t)]\psi_\epsilon(\mathbf{x}-\mathbf{X}_n(s,t))ds \\ &+ \sum_{i=1}^{n_B}\int_{L'}^L [-\mathbf{f}^{\mathrm{r}}_{n,i}(s,t)]\psi_\epsilon(\mathbf{x}-\mathbf{X}_n(s,t))ds \\ &+ \sum_{i=1}^{n_B}\int_{L'}^L \mathbf{f}^{\mathrm{r}}_{n,i}(s,t)\psi_\epsilon(\mathbf{x}-\mathbf{X}^{\mathrm{B}}_i(t))ds\Bigg] \\ &+ \sum_{i=1}^{n_B}[-\mathbf{F}^{\mathrm{B}}_i(t)]\psi_\epsilon(\mathbf{x}-\mathbf{X}^{\mathrm{B}}_i(t)).\end{aligned} \tag{11}$$

The first two terms in the force density describe the forces and torques generated by the flagella, while the last term describes the force exerted on the fluid by the cell body. To prevent contact between the flagella, as well as between the flagella and the cell body, the repulsive forces $\mathbf{f}^{\mathrm{r}}_n(s,t)$ and $\mathbf{f}^{\mathrm{r}}_{n,i}(s,t)$ are introduced as follows:

$$\begin{aligned}\mathbf{f}^{\mathrm{r}}_n(s,t) = &\sum_{\substack{n'=1\\ n'\neq n}}^{n_F}\int C\left[\max\left(1-\frac{|\mathbf{X}_n(s,t)-\mathbf{X}_{n'}(s',t)|}{d_{min}},0\right)\right. \\ &\left.\cdot\frac{\mathbf{X}_n(s,t)-\mathbf{X}_{n'}(s',t)}{|\mathbf{X}_n(s,t)-\mathbf{X}_{n'}(s',t)|}\right]ds', \\ \mathbf{f}^{\mathrm{r}}_{n,i}(s,t) = &\, C\left[\max\left(1-\frac{|\mathbf{X}_n(s,t)-\mathbf{X}^{\mathrm{B}}_i(t)|}{d_{min}},0\right)\cdot\frac{\mathbf{X}_n(s,t)-\mathbf{X}^{\mathrm{B}}_i(t)}{|\mathbf{X}_n(s,t)-\mathbf{X}^{\mathrm{B}}_i(t)|}\right].\end{aligned} \tag{12}$$

Here, the constant $d_{min}$ is the minimum distance allowed between the flagella and between the flagella and the cell body, and $C$ determines the strength of the repulsive force required to maintain this separation. The repulsive force is not applied near the motor by selecting a lower limit $L'$ for the integrals. In this study, we set $L' = 5\Delta s$. The regularized Dirac delta function $\psi_\epsilon(\mathbf{r})$ is defined as

$$\psi_\epsilon(\mathbf{r}) = \frac{15\epsilon^4}{8\pi(|\mathbf{r}|^2+\epsilon^2)^{7/2}}, \tag{13}$$

where $\epsilon$ is the regularization parameter. The blob function $\psi_\epsilon$ is a bell-shaped function with infinite support, satisfying $\int_{\mathbb{R}^3}\psi_\epsilon(\mathbf{r})d\mathbf{r} = 1$. It distributes the majority of the force and moment within a region of radius $\epsilon$ centered at the flagellum or the cell body.

The motion of the flagella and the cell body, $\mathbf{X}_n$ and $\mathbf{X}^{\mathrm{B}}_i$, respectively, is described by

$$\begin{aligned}&\frac{\partial \mathbf{X}_n}{\partial t}(s,t) = \mathbf{u}(\mathbf{X}_n(s,t),t), \qquad n = 1,\ldots,n_{\mathrm{F}}, \\ &\frac{\partial \mathbf{X}^{\mathrm{B}}_i}{\partial t}(t) = \mathbf{u}(\mathbf{X}^{\mathrm{B}}_i(t),t), \qquad i = 1,\ldots,n_{\mathrm{B}}, \\ &\frac{\partial \mathbf{D}^j_n}{\partial t}(s,t) = \mathbf{w}(\mathbf{X}_n(s,t),t)\times\mathbf{D}^j_n(s,t), \\ &\qquad\qquad j = 1,2,3 \quad \text{and} \quad n = 1,\ldots,n_{\mathrm{F}}.\end{aligned} \tag{14}$$

Here, the cell moves with the fluid velocity, while the orthonormal triads along the flagellum rotate with the angular velocity $\mathbf{w}$ of the fluid, defined as $\mathbf{w}(\mathbf{x},t) = \frac{1}{2}\nabla\times\mathbf{u}(\mathbf{x},t)$.

Each flagellar motor generates a constant torque $\tau_n$ in the normal direction $\mathbf{E}_n(t)$ to the cell surface at the motor point $\mathbf{X}_n(0,t)$ as follows:

$$\mathbf{N}_n(0,t) = -\tau_n\mathbf{E}_n(t). \tag{15}$$

Positive values of $\tau_n$ correspond to CCW rotation and negative values correspond to CW rotation. This motor torque drives flagellar rotation, enabling the cell to swim in the fluid based on the coordination of the two flagellar rotations. The parameter values used in this study are provided in Table I. The geometric parameters for the cell body and flagella were adopted from reported values for AMB-1.[18] The numerical procedure is outlined in the Appendix. Further details of the mathematical formulation, the numerical method, and their applications can be found in the relevant literature.[13,14,16,37,38]

## III. RESULTS

The swimming modes of flagellated bacteria are closely associated with their physical and geometrical properties. In this study, we first consider a cell with a single flagellum and explore how variations in the torque produced by the flagellar motor, as well as in the shape of the cell body and flagellum, affect both swimming modes and propulsion speed. Building on these insights, we develop a cell model comprising a helical cell body and two polar flagella, identify parameter regimes that reproduce the characteristic swimming behavior observed in the magnetotactic bacterium *Magnetospirillum magneticum* AMB-1, and examine the underlying mechanisms of cellular reorientation.

AMB-1 cells navigate their fluid environment to locate nutrients and avoid harmful substances. This chemotactic behavior is achieved by coordinating their two polar flagella, enabling transitions between distinct swimming modes in response to the surrounding geomagnetic field. To understand the role of magnetic guidance in their navigation, we investigate the cell dynamics in the presence of an external magnetic field, elucidating how magnetotaxis contributes to their overall locomotion strategy.

### A. A helical cell with a single polar flagellum

In this section, we begin by characterizing stable swimming behavior of a right-handed (RH) helical cell with a left-handed (LH) helical flagellum and subsequently investigate how variations in cell

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**TABLE I.** Computational and physical parameters.

| Parameters (symbol) | Value | Unit |
|---|---|---|
| Mesh width of flagellum ($\Delta s$) | 0.0302 | $\mu$m |
| Helical radius of filament ($r_f$) | 0.25 | $\mu$m |
| Helical pitch of filament ($P$) | 2.5 | $\mu$m |
| Axial length of helical filament ($L_f$) | 2.5 | $\mu$m |
| Curvilinear length of filament ($\ell_f$) | 2.985 | $\mu$m |
| Bending modulus of filament ($a_1 = a_2 = a$) | 0.003 | g $\mu$m$^3$ s$^{-2}$ |
| Twist modulus of filament ($a_3$) | 0.003 | g $\mu$m$^3$ s$^{-2}$ |
| Shear modulus ($b_1 = b_2$) | 2 | g $\mu$m$^3$ s$^{-2}$ |
| Stretch modulus ($b_3$) | 2 | g $\mu$m s$^{-2}$ |
| Length of hook ($L_h$) | $4\Delta s$ | $\mu$m |
| Bending modulus of hook ($a_{1,2}^{hook} = a^{hook} = a/25$) | 0.000 12 | g $\mu$m$^3$ s$^{-2}$ |
| Twist modulus of hook ($a_3^{hook} = a_3$) | 0.003 | g $\mu$m$^3$ s$^{-2}$ |
| Torque ($\tau$) | $-0.006 \sim 0.006$ | g $\mu$m$^2$ s$^{-2}$ |
| Number of flagella ($n_F$) | 1 or 2 | |
| Number of material points on each flagellum ($M$) | 100 | |
| Helical radius of cell body ($A$) | 0.2 | $\mu$m |
| Helical pitch of cell body ($\lambda$) | 2.5 | $\mu$m |
| Number of helical turns of cell body | 1 | |
| Thickness of cell body ($D = 2r$) | 0.5 | $\mu$m |
| Axial length of cell body ($L_b$) | 3 | $\mu$m |
| Penalty parameter between the body points ($K$) | 5.5 | g/s$^2$ |
| Penalty parameter for the flagellum to be normal to the body surface at the motor point ($\gamma$) | 20 | g/s$^2$ |
| Number of material points on the cell body ($n_B$) | 1794 | |
| Fluid viscosity ($\mu$) | $0.01 \times 10^{-4}$ | g($\mu$ms)$^{-1}$ |
| Parameter for repulsive force ($C$) | 100 | gs$^{-2}$ |
| Minimum distance between the flagella and between the flagella and the cell body ($d_{min}$) | 0.09 | $\mu$m |
| Regularization parameter ($\varepsilon$) | 0.09 | $\mu$m |
| Time step ($\Delta t$) | $4 \times 10^{-9}$ | s |

body geometry and flagellar configuration influence the resulting swimming dynamics.

We first classify the swimming modes of a cell comprising a right-handed helical cell body and a single left-handed helical flagellum as a function of the applied motor torque $\tau$, with $\tau > 0$ and $\tau < 0$ corresponding to counterclockwise (CCW) and clockwise (CW) rotations, respectively. Unless otherwise specified, the parameter values used in these simulations are listed in Table I. The torque generated by the motor is transmitted through the hook to the flagellum, inducing flagellar rotation and generating propulsion of the cell. After an initial transient period, the system reaches a stable swimming motion. Depending on the magnitude and direction of the applied torque, four distinct and stable swimming modes are observed: push, overwhirling, pull, and wrapping, see Fig. 2 (Multimedia view).

When the motor rotates CCW, there exists a critical torque, denoted as $\tau_{ccw}$, that separates the push mode from the overwhirling mode. For $0 < \tau < \tau_{ccw}$, the CCW-rotating flagellum and the counterrotating cell body together propel the bacterium forward, characterizing the push mode, see Fig. 2(a). However, when $\tau > \tau_{ccw}$, the bacterium enters the overwhirling mode, in which the flagellum undergoes large-amplitude rotation, eventually bends toward the cell body, and rotates around the cell body's axis in a steady state, as observed in Fig. 2(b). A similar transition is observed under CW motor rotation ($\tau < 0$). A critical torque, $\tau_{cw}$, separates the pull mode from the wrapping mode. For $\tau_{cw} < \tau < 0$, the CW-rotating flagellum and the resulting CCW-rotating cell body together produce backward propulsion, referred to as the pull mode [Fig. 2(c)]. For $\tau < \tau_{cw}$, the bacterium transitions to the wrapping mode in which the flagellum coils around the cell body in a corkscrew-like manner and rotates steadily, as shown in Fig. 2(d).

In polarly flagellated bacteria, which possess a single flagellum at one pole, swimming direction is defined relative to the flagellated pole. The forward run corresponds to motion away from the flagellated pole, i.e., toward the opposite, non-flagellated pole. Conversely, the backward run refers to motion toward the flagellated pole. Accordingly, the push mode underlies the forward run, with the flagellum pushing the cell body from behind, while the pull mode underlies the backward run, with the flagellum pulling the cell body from the front. Within this framework, the wrapping mode results in motion toward the flagellated pole and is thus classified as a backward

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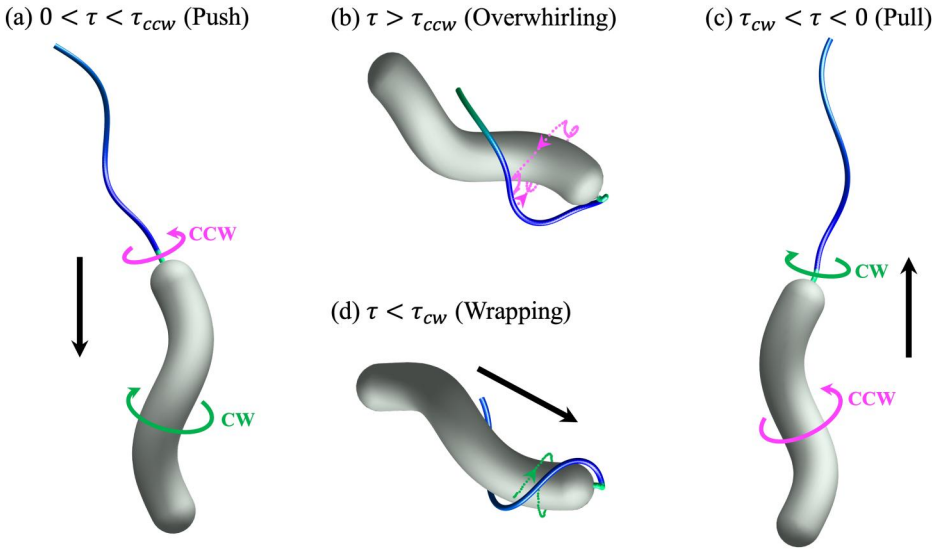



**FIG. 2.** Stable swimming modes of a polarly flagellated helical bacterium. The cell body exhibits a right-handed helical geometry, while the flagellum is intrinsically left-handed. Four distinct stable swimming modes are identified, depending on the direction of motor rotation and the magnitude of the applied torque. A positive critical torque, $\tau_{ccw}$, distinguishes the push mode (a) from overwhirling mode (b), while a negative critical torque $\tau_{cw}$ separates the pull mode (c) from the wrapping mode (d). Black arrows indicate the direction of swimming. The dotted lines in panels (b) and (d) represent a typical trajectory of a material point on each flagellum. Multimedia available online.

run. In contrast, the overwhirling mode, which arises when the motor rotates CCW, can produce both forward and backward runs depending on the magnitude of the applied torque. At moderate torque magnitudes, the cell moves backward, while at higher torques, the dynamics of the overwhirling flagellum can reverse the swimming direction, leading to forward motion. Note that the centroid of the cell in any swimming mode described here draws a helical trajectory characterized by a certain helical radius and pitch. Therefore, the directional swimming speed can be measured by tracking the overall travel distance along the axis of these helical trajectories. In general, the push, pull, and wrapping modes tend to exhibit small helical radii, whereas the overwhirling mode displays a larger helical amplitude with small superimposed oscillations.

Figure 3(a) illustrates swimming modes and corresponding speeds of a right-handed (RH) helical bacterium with a left-handed (LH) helical flagellum as functions of the applied torque. Note that negative torque values correspond to CW motor rotation, while positive values correspond to CCW rotation. Regarding swimming direction, data points marked with a cross indicate forward runs, whereas all other markers represent backward runs. The vertical dashed lines display the approximate critical torque values, $\tau_{cw}$ and $\tau_{ccw}$, which distinguish two distinct swimming modes for each direction of motor rotation. In general, within a given swimming mode, the swimming speed increases monotonically with increasing magnitude of the applied torque, except in the overwhirling regime. Within the overwhirling mode, the cell exhibits a reversal in swimming direction. Unlike other swimming modes, in which the flow generated by the flagellum is aligned with the axis of the cell body, the flagellum in the overwhirling mode undergoes a crankshafting motion—rotating around the cell body with a noticeable lateral offset. This offset, or distance between the flagellum and the cell body, depends on the applied torque. At lower torques, the flagellum remains farther from the cell body, whereas at higher torques it is positioned closer. This variation in spatial configuration influences the hydrodynamic interactions and results in distinct swimming directions.

Figure 3(b) also presents swimming modes and corresponding speeds for a left-handed (LH) helical shape, while the helical flagellum remains left-handed (LH). Compared to the RH helical cell body shown in Fig. 3(a), the LH helical cell exhibits similar trends in swimming speed and comparable critical torque thresholds. However, the

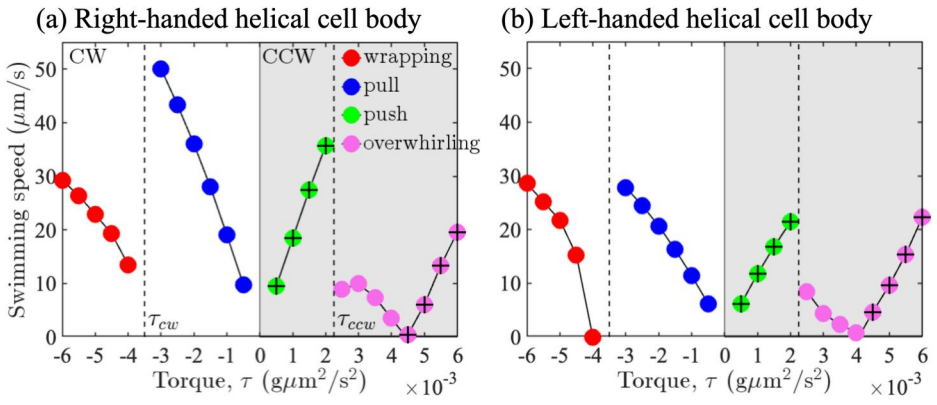



**FIG. 3.** Swimming modes and corresponding speeds of a helical bacterium with a left-handed helical flagellum as functions of the applied torque $\tau$ for two distinct cell body shapes: (a) a right-handed helical shape and (b) a left-handed helical shape. Negative torque values correspond to CW motor rotation, while positive values correspond to CCW rotation. Data points marked with a cross indicate forward runs, whereas all other markers represent backward runs. The vertical dashed lines display the approximate critical torque values, $\tau_{cw}$ and $\tau_{ccw}$, which indicate the thresholds at which transitions between two distinct swimming modes occur for each direction of motor rotation.

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swimming speeds in both push and pull modes are significantly reduced. This reduction arises because the cell body and the flagellum drive the fluid flows in opposite directions. It is worth mentioning that both LH and RH helical structures induce directional fluid flow depending on their sense of rotation: a LH helix rotating counterclockwise (CCW) or a RH helix rotating clockwise (CW) drives fluid upward, while a LH helix rotating clockwise (CW) or a RH helix rotating counterclockwise (CCW) drives fluid downward. Consequently, when the cell body and flagellum have opposite handedness, their induced flows are aligned in the same direction and reinforce each other. In contrast, when both structures share the same handedness, they generate opposing flows that partially cancel, thereby reducing propulsion. Within the wrapping mode, jamming is observed at a low torque magnitude, $\tau = -0.004\,\mathrm{g}\,\mu\mathrm{m}^2\,\mathrm{s}^{-2}$, due to the identical handedness of the cell body and the flagellum. Hydrodynamic interactions drive the flagellum to wrap around the cell body, where the two structures become stuck together without further rotation, resulting in a near-zero swimming speed. However, the swimming speeds of RH and LH helical bodies converge as the torque magnitude increases, because the thick cell body contributes little to the flow and the motion is predominantly driven by the rotating thin flagellum.

Figure 4 (Multimedia view) illustrates swimming directions and flow fields for the push and wrapping modes in both RH and LH helical cell bodies, each coupled with a left-handed helical flagellum. Figures 4(a) and 4(b) correspond to the push mode, whereas Figs. 4(c) and 4(d) correspond to the wrapping mode. In each panel, the green and red arrows, together with the indicated directions of rotation, represent the flow directions generated by the flagellum and the cell body, respectively, and the black arrow denotes the resulting swimming direction of the bacterium. In addition, each panel visualizes the flow field through streamlines and distributions of the z-directional velocity evaluated at three horizontal planes. For the RH helical body, rotations of both the flagellum and the cell body generate thrust in the same direction during both the push and wrapping modes, see Figs. 4(a) and 4(c). In contrast, for the LH helical body, these rotations produce opposing thrusts, which can result in a substantial reduction in swimming speed. For example, when $\tau = -0.004\,\mathrm{g}\,\mu\mathrm{m}^2\,\mathrm{s}^{-2}$, the torques generated by the flagellum and the cell body cancel each other, leading both structures to cease rotation and rendering the bacterium nearly stationary.

Figure 5 provides insights into how geometric parameters of a RH helical cell body affect the swimming speeds, while the flagellar geometry remains constant. The parameters considered here are (a) helical radius $A$, (b) number of helical turns $k = L_\mathrm{b}/\lambda$ with $L_\mathrm{b}$ fixed, (c) body width $D = 2r$, and (d) axial length $L_\mathrm{b}$. In each case, all the other parameters are held fixed at their default values, as listed in Table I. As the helical amplitude increases from 0.1 to 0.3 $\mu$m, the swimming speed initially increases, reaching a maximum near $A = 0.2\,\mu$m, followed by a slight decrease at higher amplitudes, see Fig. 5(a). For the number of helical turns $k$, the swimming speed increases linearly from 0 to 1, peaks at $k = 1$, after which it drops and then plateaus [see Fig. 5(b)]. The swimming speed decreases monotonically with increasing body thickness for the given range, with the highest speed observed at the smallest width of 0.4 $\mu$m and the lowest at 0.8 $\mu$m [see Fig. 5(c)]. As expected, thinner bodies experience less hydrodynamic drag, enabling more efficient propulsion. A similar decreasing trend is observed with increasing axial length [see Fig. 5(d)]. Note that Fig. 5 illustrates the trends in swimming speed when only a single parameter is varied, while all others are kept constant. Simultaneous variation of multiple parameters may further provide a map of optimal parameter conditions due to the coupled effects among the parameters.

Finally, we investigate the effect of different flagellar configurations on swimming dynamics in the push mode, under a fixed applied torque of $\tau = 0.002\,\mathrm{g}\,\mu\mathrm{m}^2\,\mathrm{s}^{-2}$ for a RH helical cell. Specifically, we consider two geometric parameters of the flagellum: the helical pitch ($P$) and the axial length ($L_\mathrm{f}$), while keeping all the other parameters fixed at the default values listed in Table I. The axial length $L_\mathrm{f}$ ranges from 1 to 6 $\mu$m and the ratio of helical pitch and helical radius, denoted by $\alpha = P/r_\mathrm{f}$, ranges from 2 to 14, where the helical radius is fixed as $r_\mathrm{f} = 0.25$. Figure 6 shows resultant representative flagellar shapes (a) and swimming speeds (b) for various combinations of $L_\mathrm{f}$ and $\alpha$. The simulation results indicate that swimming performance is highly dependent on both parameters. Four configurations producing the maximum swimming speed are highlighted with colored boxes (black, green, blue, and red) in panel (a) and corresponding colored stars in panel (b). For each axial length $L_\mathrm{f}$, swimming speed generally increases and then decreases as $\alpha$ increases. The parameter pairs that yield optimal swimming speeds appear to follow a roughly linear relationship. These results further demonstrate that neither very short nor excessively elongated flagella are optimal. Instead, intermediate values of both geometric parameters yield the most efficient propulsion, emphasizing the importance of appropriately tuning flagellar morphology to achieve enhanced motility.

### B. Bipolar helical bacteria

Amphitrichous bacteria, such as AMB-1, possess two flagella embedded at opposite poles of the cell body, and their swimming behavior and directional reversals are coordinated by the rotational states of these flagella. Each flagellum can adopt one of the four swimming modes described above; however, the combined rotational states of the two flagella give rise to a variety of distinct swimming patterns. Figures 7(a)–7(g) (Multimedia views) illustrate representative swimming motions obtained from numerical simulations of a cell with two polar flagella, constructed based on the geometry of AMB-1, in which the black curves indicate the trajectories of the cell center and the arrows represent the swimming directions.[18]

Figures 7(a)–7(d) correspond to cases in which the two flagella rotate under identical torques and adopt the same swimming mode at both poles. Under these conditions, each flagellum generates a flow of comparable magnitude but oriented in opposite directions. Due to the asymmetry of the cell body and the off-center placement of the flagellar motors relative to the helical body axis, these opposing flows do not cancel completely, resulting in the distinct swimming patterns observed in the panels (a)–(d). Among these configurations, the pull–pull mode yields the smallest net displacement, over the same simulation duration, in the direction perpendicular to the cell axis while the cell rotates as a whole.

When the flagella adopt different swimming modes, they generate flow fields that can be either aligned or oppositely directed. Aligned flows enhance swimming opposite to the induced fluid motion, whereas oppositely directed flows produce net motion dominated by the mode generating a stronger flow. Representative cases are shown in Figs. 7(e)–7(g). The wrapping–push combination (e)

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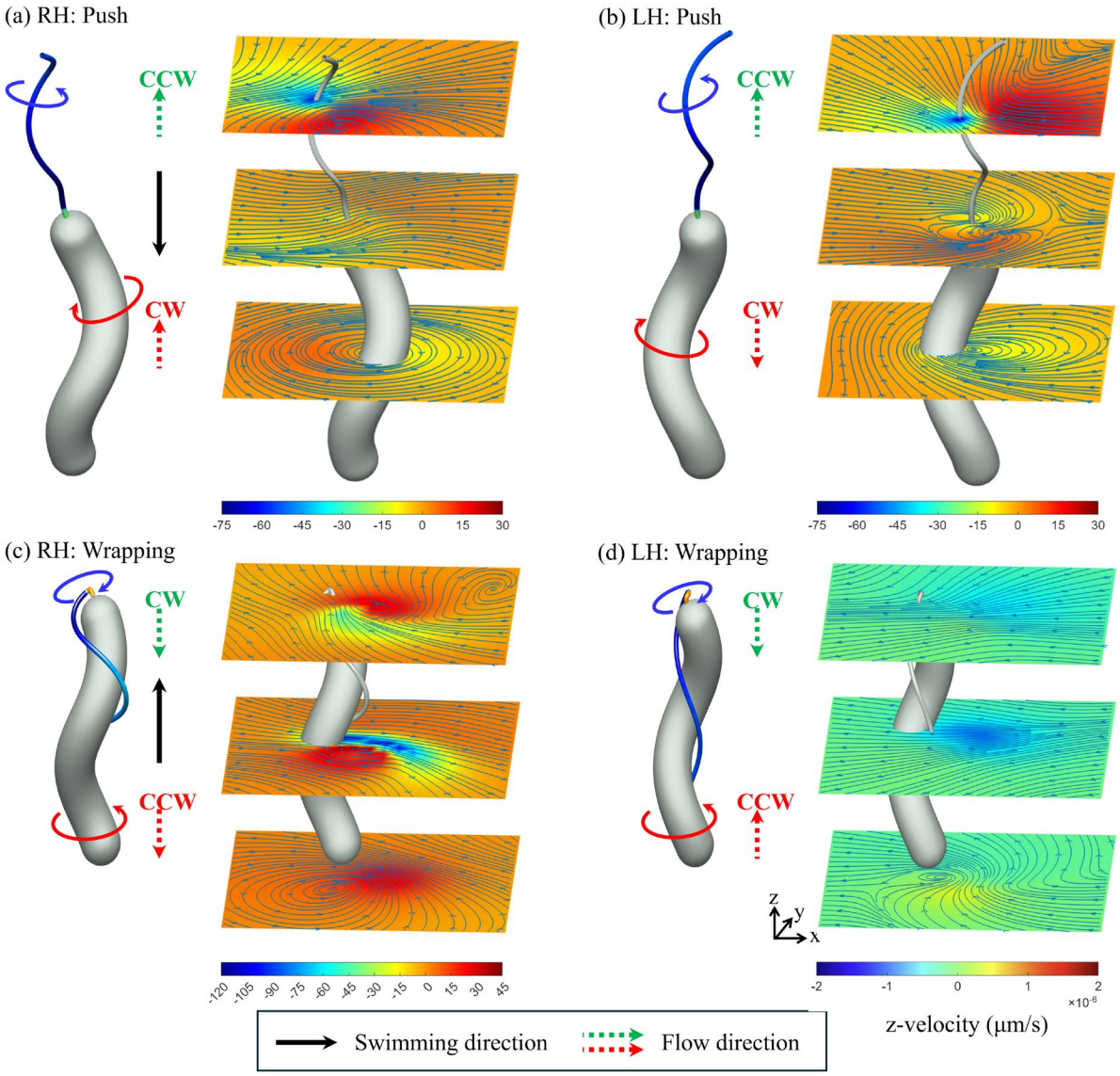



**FIG. 4.** Comparison of flow directions and flow fields during push ($\tau = 0.0015\,\mathrm{g}\,\mu\mathrm{m}^2\,\mathrm{s}^{-2}$) and wrapping ($\tau = -0.004\,\mathrm{g}\,\mu\mathrm{m}^2\,\mathrm{s}^{-2}$) configurations for two distinct cell-body morphologies. Panels (a) and (c) correspond to RH cell bodies, and panels (b) and (d) correspond to LH cell bodies. In all cases, the flagellum has a left-handed helical shape. Panels (a) and (b) show the push mode, whereas panels (c) and (d) show the wrapping mode. The green and red arrows indicate the flow directions generated by the flagellum and the cell body, respectively, and the black arrow denotes the resulting swimming direction of cells. The three horizontal planes in each panel show the streamlines and the z-velocity distributions. Multimedia available online.

consistently generates flows in the same direction, whereas the wrapping–pull combination (f) produces opposing flows. Overwhirling–push combinations (g) can result in either scenario depending on the torque magnitude of the overwhirling mode.

Figure 7(h) shows swimming speeds obtained for the 78 selected combinations of the two flagellar motor torques shown in Fig. 3(a). Five marker shapes are used in this figure. Circles indicate cases in which the two flagella retain their individual swimming modes when the torques are assigned based on the corresponding single-flagellum modes. In contrast, stars, squares, triangles, and rhombi denote cases in which one or both flagella deform into different shapes from the modes in Fig. 3(a) due to the combined effects of flagellar hydrodynamics and the asymmetric helical cell body, which typically occurs near threshold values, $\tau_{ccw}$ and $\tau_{cw}$. Among all simulations, the push–wrapping and push–pull mode pairs yield higher swimming speeds, as both flagella generate thrust in the same direction, enabling steady swimming.

Among non-circular markers, three distinct trends are observed. In the push–push cases [see the red dashed box in Fig. 7(h)], the flagellum with the higher or equal torque magnitude induces turning of the cell body, resulting in an overwhirling mode. In the overwhirling–pull

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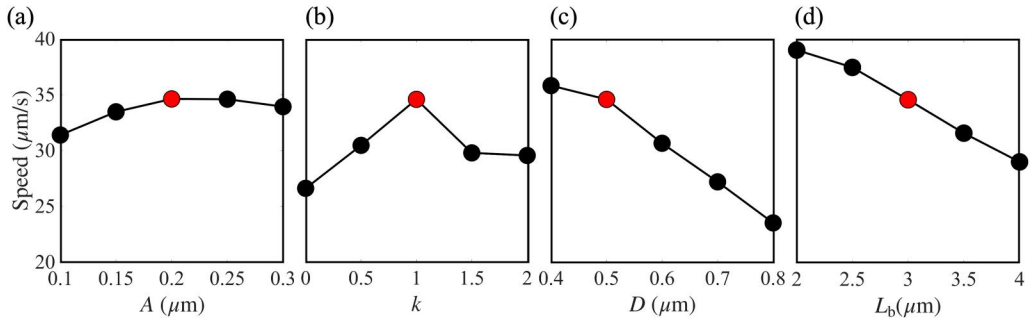



**FIG. 5.** Swimming speeds of a RH helical cell body as functions of four different geometric parameters of the cell body: (a) amplitude (helical radius) $A$, (b) number of helical turns $k$ with $L_b$ fixed, (c) width (thickness of the cell body) $D$, and (d) axial cell body length $L_b$. The default values listed in Table I are marked by red circles.

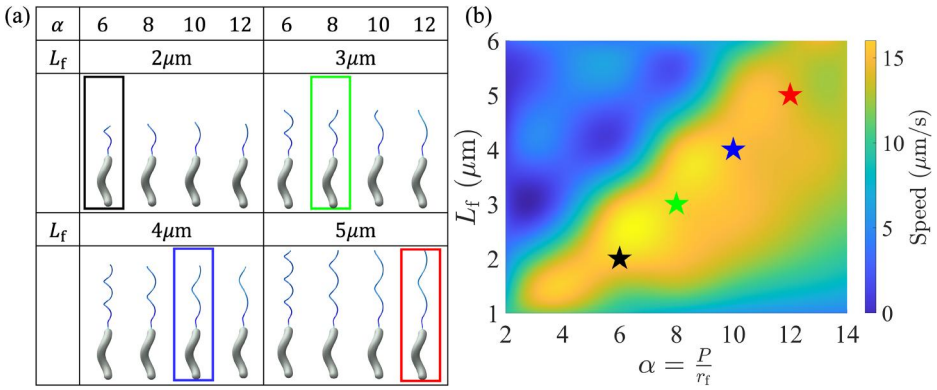



**FIG. 6.** Effect of flagellar geometry on the swimming performance of an RH helical cell body in the push mode ($\tau = 0.002$). (a) Flagellar configurations for representative combinations of axial length $L_f$ and geometric ratio $\alpha = P/r_f$, where $P$ is the helical pitch and the helical radius is fixed at $r_f = 0.25\mu$m. Four configurations are highlighted with colored boxes (black, green, blue, and red), each corresponding to the value of $\alpha$ that yields the maximum swimming speed for the given $L_f$. (b) Contour plot of the swimming speed as a function of $L_f$ and $\alpha$. Four stars correspond to the four boxes with matching colors in (a).

cases [see the blue dashed box in Fig. 7(h)], the combined effect of the two modes enhances cell body rotation, causing the pull-mode flagellum to transition into a wrapped configuration. In the pull–pull or pull–wrapping cases [see the black dashed box in Fig. 7(h)], the pull-mode flagellum bends toward the cell body, and the two flagella rotate around the cell body in opposite directions, hindering smooth rotation or effectively locking one flagellum. The right panel of Fig. 7(h) shows representative configurations corresponding to four non-circular markers.

Amphitrichous bacterium AMB-1 experiences runs, pauses, and reversals arising from the hydrodynamic interplay between the two flagellar motor states and the helical cell body.[18] Figure 8 (Multimedia view) presents a representative sequence of cell's behaviors resulting from changes in the direction of motor rotation and the magnitude of the applied torque. The selected sequence of swimming-mode pairs reflects behaviors observed experimentally in AMB-1 bacteria.[18,33] Figure 8(a) shows the time evolution of the torque applied to each flagellum together with the corresponding swimming modes, while Fig. 8(b) displays the resulting cell trajectory. Note that we include "pause" events, during which the flagellar motor stops, as observed experimentally.[18]

The bacterium initially undergoes an almost straight run via a push–wrapping combination (I), followed by a pause–pause state (II) in which both flagellar motors are temporarily inactive, resulting in minimal cell movement. This is succeeded by a pull–pull configuration (III), during which the cell moves perpendicular to its cell axis. The subsequent transition to a pull–wrapping combination (IV) is accompanied by a change in swimming direction. During the following pull–pause (V) and wrapping–pause (VI) intervals, propulsion occurs in only one direction, as the same flagellum remains active in either the pull or wrapping mode. The sequence concludes with a wrapping–wrapping configuration (VII), in which both flagella operate in the same mode. This ordered progression of mode transitions corresponds directly to the time evolution of the motor torques shown in Fig. 8(a). The resulting trajectories of the cell center are displayed in different colors corresponding to mode transitions in Fig. 8(b), together with configurations captured at specific times (marked with dots) for each mode pair.

A distinct feature arises during the transition from wrapping to pull: the flagellum wrapping around the cell body must first unwind before the pull mode can be established, which requires a short pause of approximately 0.02 s or longer. In contrast, other mode transitions proceed smoothly without comparable pauses, as the rotational states of the flagella change. Large changes in swimming direction also occur during flagellar wrapping or unwrapping events, similar to observations in other bacteria such as *P. putida* and *C. jejuni*.[5,6]

### C. Effect of the external magnetic field on bacterial motility

*M. magneticum* AMB-1 contains a chain of magnetosomes within the cell body, which conveys a magnetic moment to the cell

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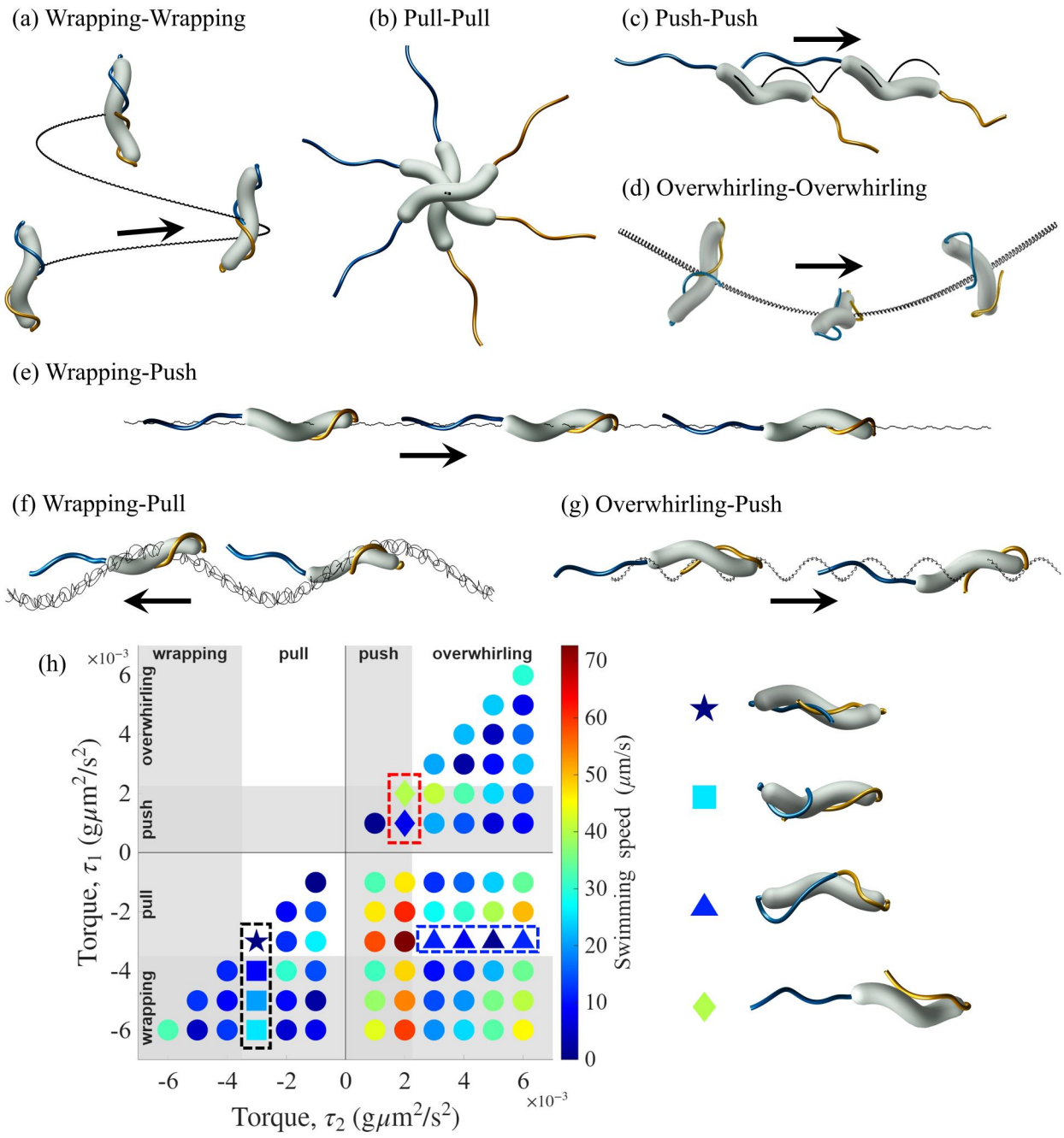



**FIG. 7.** Representative swimming-mode combinations in an amphitrichous bacterium (a)–(g) and swimming speeds for various combinations of modes (h). The motor torque used for each mode in panels (a)–(g) is as follows: wrapping ($\tau = -0.005$), pull ($\tau = -0.0015$), push ($\tau = 0.0015$), or overwhirling ($\tau = 0.005$). Panels (a)–(d) illustrate swimming patterns when both motors rotate in the same mode with identical torque magnitudes, whereas panels (e)–(g) present representative trajectories for cases in which the two motors operate in distinct swimming modes. All arrows indicate the swimming direction of the cells. The black curves indicate the trajectories of the cell center. For visualization purposes, the bacterium is rendered at a reduced scale. Panel (h) displays swimming speeds resulting from 78 pairs of motor torques, as specified in Fig. 3(a). Symmetric data points in the top-left region are omitted for clarity. Circles indicate cases where both flagella retain their single-flagellum swimming modes under the assigned torques. Stars, squares, triangles, and rhombi enclosed by dashed boxes indicate cases where the flagella deform into different shapes due to hydrodynamic interaction with the asymmetric helical cell, typically occurring near threshold values. The right panel of Fig. 7(h) shows representative configurations corresponding to four non-circular markers. Multimedia available online.

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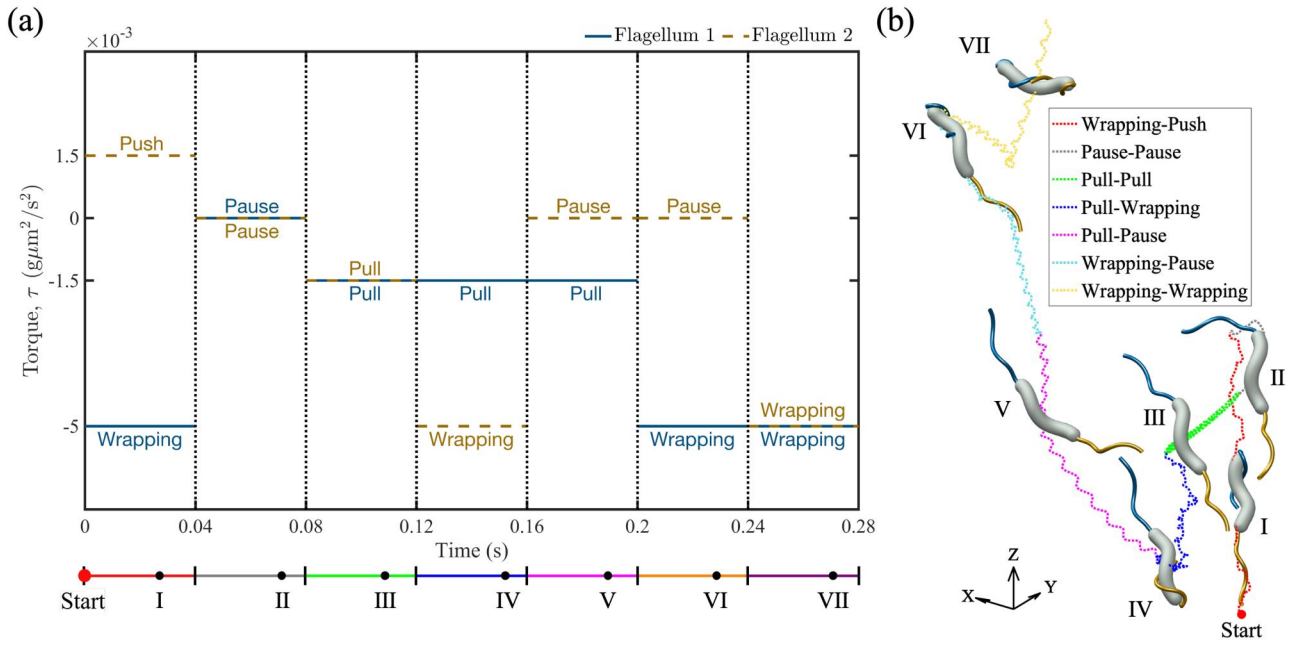



**FIG. 8.** Exemplary sequence of swimming-mode combinations of AMB-1 and the corresponding trajectory of the cell center. Panel (a) shows time evolution of torques applied to each flagellum, producing runs, pauses, and reversals. Panel (b) displays the resulting swimming trajectory, with different colors indicating various pairs of swimming modes of the two flagella. Cell configurations at seven time points [marked with dots at the bottom of (a)] for each mode pair are captured at a reduced scale for clarity. Multimedia available online.

and enables orientation in an external magnetic field. This internal magnetic compass allows magnetotactic bacteria to align passively and swim along geomagnetic field lines, a mechanism proposed to facilitate their navigation toward the oxic–anoxic transition zone, their preferred habitat[18,29,33]

To investigate the magnetic field-induced motion of AMB-1, we account for the coupled effects of magnetic and hydrodynamic interactions acting on the cell. The cell model includes a chain of magnetosomes (see Fig. 1), which generates a permanent magnetic moment $\mathbf{m}$ aligned with the helical axis of the cell body. The magnitude of the magnetic moment is set to $\|\mathbf{m}\| = 1.26 \times 10^{-15}\,\mathrm{A\,m^2}$, consistent with experimental measurements for AMB-1.[46] Since the magnetosome chain is fixed within the cell body, the magnetic moment is assumed to rotate rigidly with the body. In the presence of an external magnetic field $\mathbf{B}$, the magnetic dipole experiences a torque that tends to align $\mathbf{m}$ with $\mathbf{B}$, given by

$$\mathbf{n}_\mathrm{m} = \mathbf{m} \times \mathbf{B}. \tag{16}$$

This magnetic torque drives rotation of the cell body and, under a uniform magnetic field, constitutes the sole contribution to the externally applied rotational torque. The motion of AMB-1 is therefore governed by the balance between the hydrodynamic torques generated by flagellar rotation and the magnetic torque induced by the external field. Accordingly, the magnetic torque is incorporated into the torque balance equation to compute the translational velocity, rotational dynamics, and magnetic alignment of the cell body across a range of magnetic field strengths and motor torque conditions.

To validate our model's ability to capture the magnetic steering behavior of AMB-1, we consider a cell swimming in the push-wrapping mode, which exhibits nearly straight-line motion in the absence of an external magnetic field, and examine its response to time-varying magnetic fields. In this test, the direction of the external magnetic field is continuously modulated such that its orientation traces a circle, a square, and a triangle over time. The magnetic field strength is fixed at $\|\mathbf{B}\| = 1\mathrm{mT}$, ensuring that the magnetic torque dominates hydrodynamic torques generated by the flagella. The resulting trajectories are shown in Fig. 9 (Multimedia view). In all cases, the bacterium aligns its magnetic moment with the instantaneous direction of the external field and follows the prescribed geometric paths. Smooth rotation of the field, as in the circular pattern, yields correspondingly a smooth trajectory, whereas square and triangular patterns involve abrupt directional changes that induce rapid reorientation of the cell body. These results demonstrate that the magnetic torque is sufficient to overcome hydrodynamic resistance during sharp turns and confirm that the model accurately captures the coupling between the magnetic moment and the external field, enabling reliable tracking of time-dependent magnetic cues.

To examine how an external magnetic field modifies swimming behavior, we generate a sequence of runs and tumbles under varying magnetic field strengths, see Fig. 10 (Multimedia view). The magnetic field $\mathbf{B}$ is oriented parallel to the $z$-axis, and the same torque sequence is applied across different field strengths while all the other parameters are held fixed. Figures 10(a) and 10(b) show time evolution of the torques applied to each flagellum and the resulting swimming trajectory in the absence of an external magnetic field. Figures 10(c)–10(f) show swimming trajectories obtained using the same torque evolution as in Fig. 10(a) under increasing magnetic field strengths: (c) $\|\mathbf{B}\| = 0.1\,\mathrm{mT}$, (d)

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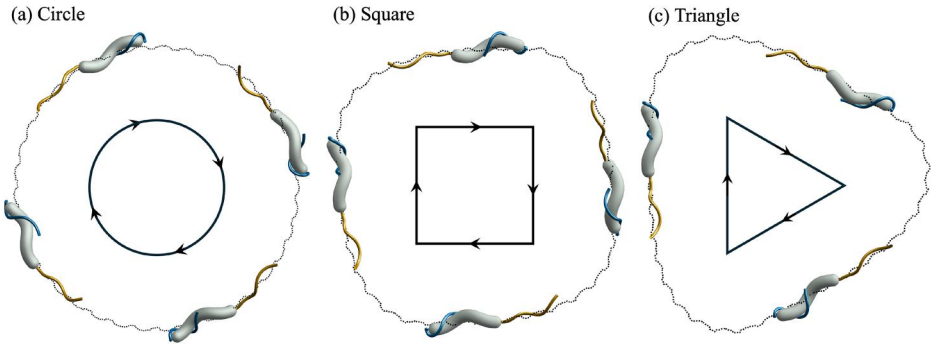



**FIG. 9.** Trajectories of a cell when the direction of the external magnetic field traces (a) a circle, (b) a square, and (c) a triangle over time, as indicated by the arrows. The magnetic field strength is fixed at $\| \mathbf{B} \| = 1\,\mathrm{mT}$. The cell swims in push–wrapping mode. For clarity, the cell is rendered at a reduced size, and black dotted lines denote the trajectory of the cell center. Multimedia available online.

$\|\mathbf{B}\| = 0.2\,\mathrm{mT}$, (e) $\|\mathbf{B}\| = 0.4\,\mathrm{mT}$, and (f) $\|\mathbf{B}\| = 1\,\mathrm{mT}$. In the absence of a magnetic field ($\|\mathbf{B}\| = 0\,\mathrm{mT}$), the cell follows its characteristic trajectory determined solely by hydrodynamic interactions associated with coordinated flagellar rotation. When an external magnetic field is applied, magnetic and hydrodynamic torques compete to set the cell orientation. As the field strength increases, swimming trajectories are increasingly deflected toward the direction of the applied magnetic field, most prominently during forward and backward runs. Moreover, in the pull–pull mode, the cell exhibits enhanced lateral motion, with trajectories generally perpendicular to the cell axis which remains closely aligned with the direction of the external field. Thus, while the intrinsic flagellar modes remain unchanged, the magnetic field progressively constrains the cell orientation and enforces a preferred swimming direction. This effect is further quantified in Fig. 10(g), which shows the time evolution of the angle between the magnetic moment **m** and the external magnetic field **B**. As expected, this angle decreases with increasing $\|\mathbf{B}\|$, reflecting the progressive alignment of the cell axis with the external field.

## IV. SUMMARY AND DISCUSSION

Motivated by amphitrichous bacteria such as *M. magneticum* AMB-1, which features a helical cell body and two polar flagella, this study investigates how coordination between the two flagella and the helical body geometry together affect the swimming motility. We further examine how the internal chain of magnetosomes influences swimming orientation under external magnetic fields. To this end, we develop a computational cell model based on the real geometry of AMB-1 to investigate the swimming dynamics. The model organism swims in an incompressible viscous fluid, with its motion determined under force- and torque-free conditions generated by the flagellar motors.

The swimming efficiency of a cell model with a single flagellum is strongly influenced by the relative handedness of the cell body and the flagellum. When the two structures possess opposite handedness, the rotationally induced flows reinforce each other, resulting in enhanced propulsion speeds. In contrast, identical handedness generates opposing flow fields that reduce propulsion and, in some cases, nearly suppress net motion. These findings emphasize that geometric chirality plays an important role in controlling propulsion efficiency in the model. In addition, geometric variations of the helical cell body further modulate swimming performance. Changes in helical amplitude, number of helical turns, body thickness, and axial length influence propulsion through their effect on hydrodynamic drag and flow generation. Optimal swimming occurs at intermediate values of helical amplitude and number of turns, whereas increased body thickness or axial length leads to a monotonic reduction in swimming speed. Similar trends are observed for flagellar geometry, and intermediate values of flagellar length and pitch yield maximal propulsion.

For a helical bacterium with two polar flagella, the overall swimming behavior is determined by the combination of swimming modes exhibited by each flagellum. The swimming speed varies depending on the torque magnitude associated with each mode pair. Each combination produces a distinct and stable helical trajectory, characterized by a specific pitch and radius, ranging from nearly zero—corresponding to a straight run—to large values. These results demonstrate that the interplay between flagellar mode selection and torque magnitude governs both the directionality and the spatial characteristics of the bacterium's motion. When the cell undergoes a sequence of different mode pairs consecutively, the cell runs and tumbles due to mode transitions. It is worth mentioning that a pause is required only during the transition from wrapping to pull, corresponding to the unwinding of the wrapped flagellum along the cell body, a phenomenon also observed in rod-shaped cells.[13] Moreover, simulation results indicate that large changes in swimming direction occur during flagellar wrapping and unwrapping events.

The effect of external magnetic fields is incorporated via a magnetic torque acting on the cell body. Our simulations show that magnetic and hydrodynamic torques compete to determine cell orientation. At weak field strengths, both contributions are comparable, whereas stronger fields increasingly constrain orientation and align swimming trajectories with the applied magnetic field. Importantly, the intrinsic sequence of flagellar modes remains unchanged in the presence of a magnetic field, indicating that magnetic fields primarily bias orientation without altering propulsion mechanisms.

Experimental observations indicate that each flagellum of AMB-1 can adopt either a parachute form, in which the flagellum bends toward the cell body, or a tuft form, in which the flagellum extends outward from the cell pole and generates propulsion by pushing or pulling depending on the motor rotation direction.[18,47] In comparison with our computational model, the parachute configuration could correspond to either the wrapping or overwhirling mode, which remains uncertain. However, our simulations suggest that the wrapping mode is more consistent, as the push–wrapping mode pair always generates flows in the same direction, resulting in more efficient propulsion (see Fig. 7). However, another possibility arises when the cell adopts the push–push mode with high or equal torque magnitudes close to the threshold. In this case, both flagella generate thrust toward the helical cell body, causing the body to turn and leading the

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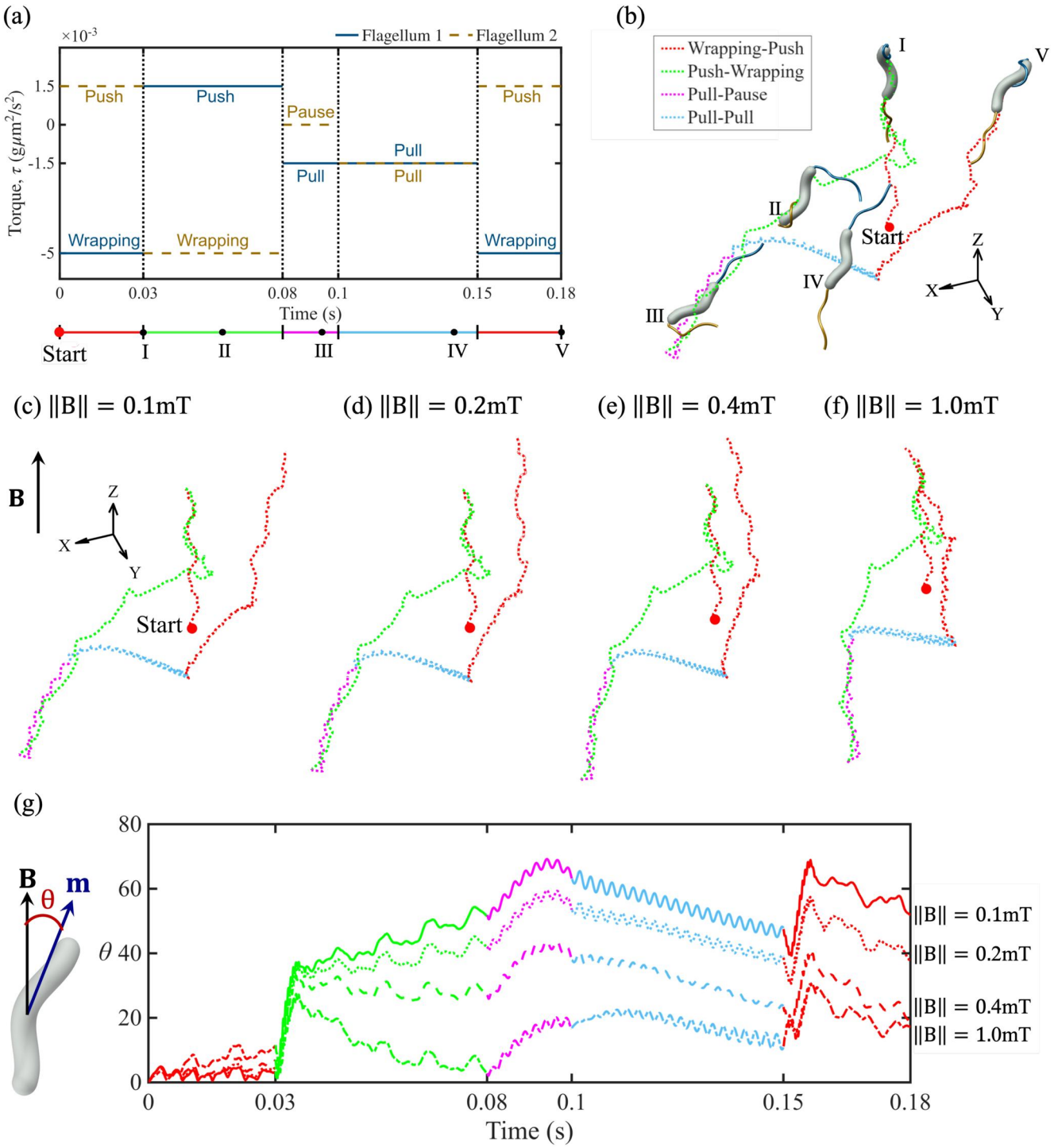



**FIG. 10.** Influence of the external magnetic field on bacterial swimming behavior. Panel (a) shows time evolution of torque values for each flagellum to produce runs and tumbles. Panel (b) shows the resulting swimming trajectory in the absence of an external magnetic field, with different colors corresponding to different pairs of swimming modes of the two flagella. Cell configurations are shown at five time points and rendered at reduced scale for visualization. Panels (c)–(f) show swimming trajectories obtained using the same torque evolution as in panel (a) under various external magnetic field strengths: (c) $\| \mathbf{B} \| = 0.1$ mT, (d) $\| \mathbf{B} \| = 0.2$ mT, (e) $\| \mathbf{B} \| = 0.4$ mT, and (f) $\| \mathbf{B} \| = 1$ mT. Panel (g) presents time evolution of the angle between the magnetic moment $\mathbf{m}$ and the external magnetic field $\mathbf{B}$. As the field strength increases, this angle decreases, indicating stronger alignment of the magnetic moment with the external field. Multimedia available online.

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higher-torque flagellum to bend toward the cell body and deform into an overwhirling configuration. This state may also be interpreted as a parachute-tuft case. Further investigation is required to definitively identify the parachute-tuft mode observed in the experimental data. Another interesting direction for future work is to examine the swimming dynamics of amphitrichous bacteria with multiple flagella at each pole, such as *Spirillum volutans*.[48,49] The presence of multiple flagella introduces additional complexity, as overall motion depends on the coordinated activity and mode selection of all flagella, potentially giving rise to new swimming patterns and emergent behaviors not observed in the case with two flagella.

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## AUTHOR DECLARATIONS

### Conflict of Interest

The authors have no conflicts to disclose.

### Author Contributions

**Sangkwon Kim:** Conceptualization (equal); Data curation (equal); Funding acquisition (equal); Investigation (equal); Methodology (equal); Writing – original draft (equal); Writing – review & editing (equal). **Wanho Lee:** Conceptualization (equal); Data curation (equal); Funding acquisition (equal); Investigation (equal); Methodology (equal); Writing – review & editing (supporting). **Yongsam Kim:** Conceptualization (equal); Data curation (supporting); Funding acquisition (equal); Investigation (supporting); Methodology (equal); Writing – review & editing (supporting). **Sookkyung Lim:** Conceptualization (equal); Data curation (equal); Funding acquisition (equal); Investigation (equal); Methodology (equal); Supervision (lead); Visualization (supporting); Writing – original draft (equal); Writing – review & editing (equal).

## DATA AVAILABILITY

The data that support the findings of this study are available within the article.

## APPENDIX: NUMERICAL PROCEDURE FOR THE COUPLED FLUID–STRUCTURE INTERACTION EQUATIONS

This Appendix outlines the numerical procedure used to advance the coupled fluid–structure system described in Sec. II in time. We first present analytic solutions for the rigid cell body equilibrium equations and the regularized fundamental solutions of the incompressible Stokes equations. We then describe the time integration of the Lagrangian positions and orthonormal triads using the computed linear and angular velocities.

### 1. Rigid body equilibrium solution

Given $\mathbf{X}_i^{\mathrm{B}}(t)$, $\mathbf{f}^{\mathrm{B}}(t)$, and $\mathbf{n}^{\mathrm{B}}(t)$ at time $t$, substituting $\hat{\mathbf{X}}_i^{\mathrm{B}} = \mathbf{T} + \mathcal{R}\,\mathbf{Z}_i^{\mathrm{B}}$ into Eq. (1), and using $\sum_i \mathbf{Z}_i^{\mathrm{B}} = \mathbf{0}$ and $(\mathcal{R}\mathbf{Z}_i^{\mathrm{B}}) \times (\mathcal{R}\mathbf{Z}_i^{\mathrm{B}}) = \mathbf{0}$, the system of Eqs. (1)–(3) reduces to

$$\mathbf{0} = \mathbf{f}^{\mathrm{B}}(t) + K \sum_{i=1}^{n_{\mathrm{B}}} (\mathbf{X}_i^{\mathrm{B}}(t) - \mathbf{T}(t)), \tag{A1}$$

$$\mathbf{0} = \mathbf{n}^{\mathrm{B}}(t) + K \sum_{i=1}^{n_{\mathrm{B}}} (\mathcal{R}\,\mathbf{Z}_i^{\mathrm{B}}) \times \mathbf{X}_i^{\mathrm{B}}(t). \tag{A2}$$

Equation (A1) gives the translation $\mathbf{T}(t)$ directly as

$$\mathbf{T}(t) = \frac{1}{K\,n_{\mathrm{B}}}\,\mathbf{f}^{\mathrm{B}}(t) + \frac{1}{n_{\mathrm{B}}} \sum_{i=1}^{n_{\mathrm{B}}} \mathbf{X}_i^{\mathrm{B}}(t). \tag{A3}$$

An approximate solution for $\mathcal{R}(t)$ can be obtained analytically from Eq. (A2) and is given by[38]

$$\mathcal{R}(t) = (I + K^{-1}\Xi)\mathcal{R}_0, \tag{A4}$$

where $\mathcal{R}_0 = (\mathcal{Q}^{\mathrm{T}}\mathcal{Q})^{1/2}\,\mathcal{Q}^{-1}$ and $\mathcal{Q} = \sum_{i=1}^{n_{\mathrm{B}}} \mathbf{Z}_i^{\mathrm{B}} (\mathbf{X}_i^{\mathrm{B}})^{\mathrm{T}}$. Here, $\Xi$ is an antisymmetric matrix and satisfies $\Xi = \sigma^{\mathrm{T}}(\sigma\Xi\sigma^{\mathrm{T}})\sigma$, where $\sigma$ is an orthogonal matrix composed of the eigenvectors of $\mathcal{R}_0\mathcal{Q}$.

### 2. Fundamental solutions of the regularized Stokes equations

Consider a single point force $\mathbf{f}_o$ and a single point torque $\mathbf{m}_o$ applied at $\mathbf{X}_o$. The body force then reads

$$\mathbf{g}(\mathbf{x}) = \mathbf{f}_o\,\psi_\epsilon(\mathbf{x} - \mathbf{X}_o) + \frac{1}{2}\,\nabla \times \left[\mathbf{m}_o\,\psi_\epsilon(\mathbf{x} - \mathbf{X}_o)\right]. \tag{A5}$$

Now, define two functions $G_\epsilon(\mathbf{x})$ and $B_\epsilon(\mathbf{x})$ as the solutions of $\Delta G_\epsilon = \psi_\epsilon$ and $\Delta B_\epsilon = G_\epsilon$, where $G_\epsilon$ is the regularized Green's function and $B_\epsilon$ is the corresponding solution to the biharmonic equation. Then, the regularized fundamental solutions of the incompressible Stokes Eq. (10) and Eq. (A5) for the local linear and angular fluid velocities can be written as follows:[37]

$$\mu\,\mathbf{u}(\mathbf{x}) = (\mathbf{f}_o \cdot \nabla)\,\nabla B_\epsilon(\mathbf{r}) - \mathbf{f}_o\,G_\epsilon(\mathbf{r}) + \frac{1}{2}\,\mathbf{m}_o \times \nabla G_\epsilon(\mathbf{r}), \tag{A6}$$

$$\mu\,\mathbf{w}(\mathbf{x}) = \frac{1}{2}\,\mathbf{f}_o \times \nabla G_\epsilon(\mathbf{r}) + \frac{1}{4}\,\mathbf{m}_o\,\psi_\epsilon(\mathbf{r}) - \frac{1}{4}\,(\mathbf{m}_o \cdot \nabla)\,\nabla G_\epsilon(\mathbf{r}), \tag{A7}$$

where $\mathbf{r} = \mathbf{x} - \mathbf{X}_o$. By the linearity of Eq. (10), the fluid velocity and angular velocity at any point are obtained by summing the contributions given by Eqs. (A6) and (A7) over all point forces and point torques appearing in Eq. (11).

### 3. Discretization and time integration

We discretize the flagellar centerline and the cell body. For each flagellum, the centerline $\mathbf{X}$ is discretized into $M$ Lagrangian

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markers with uniform spacing $\Delta s$. Let $\mathbf{X}_m^k$ denote the position of the $m$th flagellar marker at time $t = k\triangle t$, where $\triangle t$ is the time step and $m = 1, \ldots, M$. The same discretization is used for the other Lagrangian variables $\mathbf{D}^1$, $\mathbf{D}^2$, $\mathbf{D}^3$, $\mathbf{F}$, $\mathbf{N}$, $\mathbf{f}$, and $\mathbf{n}$. Let $\mathbf{D}_m^{j,k}$ denote the orthonormal triads associated with $\mathbf{X}_m^k$, where $j = 1, 2, 3$. The cell body surfaces $\hat{\mathbf{X}}_i^{\mathrm{B}}$ and $\mathbf{X}_i^{\mathrm{B}}$ are discretized using $n_{\mathrm{B}}$ Lagrangian markers, where $i = 1, \ldots, n_{\mathrm{B}}$.

Starting from the current cell body configurations $\mathbf{X}_i^{\mathrm{B},k}$ and $\hat{\mathbf{X}}_i^{\mathrm{B},k}$ at time $t = k\triangle t$, we proceed as follows:

(1) Compute the internal forces and moments, $\mathbf{F}_{m+\frac{1}{2}}$ and $\mathbf{N}_{m+\frac{1}{2}}$, transmitted across the flagellar cross sections at half-integer indices by discretizing Eq. (9).
(2) Compute the force and torque densities, $\mathbf{f}_m$ and $\mathbf{n}_m$, by discretizing Eq. (6), together with repulsive forces and the force acting on the fluid by the cell body through Eq. (11).
(3) Compute $\mathbf{f}^{\mathrm{B}}$ and $\mathbf{n}^{\mathrm{B}}$ as the sums of all forces and torques applied to the cell body, respectively, excluding the forces arising from the coupling springs.
(4) Given $\mathbf{f}^{\mathrm{B}}$, $\mathbf{n}^{\mathrm{B}}$, and $\mathbf{X}_i^{\mathrm{B},k}$, determine the translation and rotation $(\mathbf{T}, \mathcal{R})$ using Eqs. (A3) and (A4), and then compute $\hat{\mathbf{X}}_i^{\mathrm{B},k+1}$ and the penalty force $\mathbf{F}_i^{\mathrm{B}}$ from Eq. (1).
(5) Sum all contributions in Eq. (11) at each Lagrangian material point through Eqs. (A6) and (A7) to obtain the local fluid velocity $\mathbf{u}$ and angular velocity $\mathbf{w}$.
(6) Update the positions of the flagella and the cell body using the local fluid velocity as follows:

$$\begin{aligned} \mathbf{X}_m^{k+1} &= \mathbf{X}_m^k + \Delta t\, \mathbf{u}\left(\mathbf{X}_m^k\right), \qquad m = 1, \ldots, M, \\ \mathbf{X}_i^{\mathrm{B},k+1} &= \mathbf{X}_i^{\mathrm{B},k} + \Delta t\, \mathbf{u}\left(\mathbf{X}_i^{\mathrm{B},k}\right), \qquad i = 1, \ldots, n_{\mathrm{B}}. \end{aligned} \tag{A8}$$

(7) Update the orthonormal triads using the local angular fluid velocity as follows:

$$\mathbf{D}_m^{j,k+1} = \mathfrak{R}\left(\frac{\mathbf{w}(\mathbf{X}_m^k)}{|\mathbf{w}(\mathbf{X}_m^k)|}, |\mathbf{w}(\mathbf{X}_m^k)|\triangle t\right)\mathbf{D}_m^{j,k}, \tag{A9}$$

where $j = 1, 2, 3$, $m = 1, \ldots, M$, and $\mathfrak{R}(\mathbf{e}, \theta)$ is the orthogonal matrix describing a rotation through an angle $\theta$ about the unit vector $\mathbf{e}$, given by

$$\mathfrak{R}(\mathbf{e}, \theta) = (\cos\theta)\, I + (1 - \cos\theta)\, \mathbf{e}\,\mathbf{e}^{\mathrm{T}} + (\sin\theta)\, [\mathbf{e}], \tag{A10}$$

where $[\mathbf{e}]$ is the antisymmetric $3 \times 3$ matrix defined by $[\mathbf{e}]\mathbf{v} = \mathbf{e} \times \mathbf{v}$ for any vector $\mathbf{v}$.

## REFERENCES


[1]H. Berg and L. Turner, "Movement of microorganisms in viscous environments," Nature **278**, 349–351 (1979).
[2]H. Berg, "Motile behavior of bacteria," Phys. Today **53**(1), 24–29 (2000).
[3]H. Berg, "The rotary motor of bacterial flagella," Annu. Rev. Biochem. **72**, 19–54 (2003).
[4]B. Chaban, H. V. Hughes, and M. Beeby, "The flagellum in bacterial pathogens: For motility and a whole lot more," Semin. Cell. Dev. Biol. **46**, 91–103 (2015).
[5]M. Hintsche, V. Waljor, R. Großmann, M. J. K. "uhn, K. M. Thormann, F. Peruani, and C. Beta, "A polar bundle of flagella can drive bacterial swimming by pushing, pulling, or coiling around the cell body," Sci. Rep. **7**, 16771 (2017).
[6]E. J. Cohen, D. Nakane, Y. Kabata, D. R. Hendrixson, T. Nishizaka, and M. Beeby, "Campylobacter jejuni motility integrates specialized cell shape, flagellar filament, and motor to coordinate action of its opposed flagella," PLoS Pathog. **16**, e1008620 (2020).
[7]K. Son, J. S. Guasto, and R. Stocker, "Bacteria can exploit a flagellar buckling instability to change direction," Nat. Phys. **9**, 494–498 (2013).
[8]E. P. Greenberg and E. Canale-Parola, "Motility of flagellated bacteria in viscous environments," J. Bacteriol. **132**, 356–358 (1977).
[9]G. E. Kaiser and R. N. Doetsch, "Letter: Enhanced translational motion of Leptospira in viscous environments," Nature **255**, 656–657 (1975).
[10]R. E. Goldstein, A. Goriely, G. Huber, and C. W. Wolgemuth, "Bistable helices," Phys. Rev. Lett. **84**, 1631–1634 (2000).
[11]W. Schneider and R. Doetsch, "Effect of viscosity on bacterial motility," J. Bacteriol. **117**, 696–701 (1974).
[12]Y. Park, Y. Kim, and S. Lim, "Locomotion of a single-flagellated bacterium," J. Fluid Mech. **859**, 586–612 (2019).
[13]J. Park, Y. Kim, W. Lee, and S. Lim, "Modeling of lophotrichous bacteria reveals key factors for swimming reorientation," Sci. Rep. **12**, 6482 (2022).
[14]W. Lee, Y. Kim, and S. Lim, "Bio-inspired in silico microswimmer: Run and tumble kinematics," Phys. Fluids **35**, 031908 (2023).
[15]D. Kim, Y. Kim, and S. Lim, "Effects of swimming environment on bacterial motility," Phys. Fluids **34**, 031907 (2022).
[16]J. Park, Y. Kim, W. Lee, V. Pfeifer, V. Muraveva, C. Beta, and S. Lim, "Bundling instability of lophotrichous bacteria," Phys. Fluids **36**, 101917 (2024).
[17]R. Macnab, "How bacteria assemble flagella," Annu. Rev. Microbiol. **57**, 77–100 (2003).
[18]D. Murat, M. H'erisse, L. Espinosa, A. Bossa, F. Alberto, and L. Wu, "Opposite and coordinated rotation of amphitrichous flagella governs oriented swimming and reversals in a magnetotactic spirillum," J. Bacteriol. **197**, 3275–3282 (2015).
[19]F. Samatey, H. Matsunami, K. Imada, S. Nagashima, T. Shaikh, D. Thomas, J. Chen, J. Derosier, A. Kitao, and K. Namba, "Structure of the bacterial flagellar hook and implication for the molecular universal joint mechanism," Nature **431**, 1062–1068 (2004).
[20]G. Kirchhoff, "Über das Gleichgewicht und die bewegung eines unendlich dünnen elastischen stabes," J. Reine Angew. Math. **56**, 285–313 (1859).
[21]A. E. Love, *A Treatise on the Mathematical Theory of Elasticity* (Cambridge University Press, Cambridge, 1892).
[22]E. H. Dill, "Kirchhoff's theory of rods," Arch. Hist. Exact Sci. **44**, 1–23 (1992).
[23]H. Matsunami, C. Barker, Y. Yoon, M. Wolf, and F. A. Samatey, "Complete structure of the bacterial flagellar hook reveals extensive set of stabilizing interactions," Nat. Commun. **7**, 13425 (2016).
[24]T. Kato, F. Makino, T. Miyata, P. Horváth, and K. Namba, "Structure of the native supercoiled flagellar hook as a universal joint," Nat. Commun. **10**, 5295 (2019).
[25]H. Berg and D. Brown, "Chemotaxis in *Escherichia coli* analysed by three-dimensional tracking," Nature **239**, 500–504 (1972).
[26]M. J. Kühn, F. K. Schmidt, B. Eckhardt, and K. M. Thormann, "Bacteria exploit a polymorphic instability of the flagellar filament to escape from traps," Proc. Natl. Acad. Sci. U. S. A. **114**, 6340–6345 (2017).
[27]J. Errington and L. Aart, "Bacillus subtilis: Model organism for cellular development, and industrial workhorse," Microbiology **166**, 425–427 (2020).
[28]T. Bartlett, B. Bratton, A. Duvshani, A. Miguel, Y. Sheng, N. Martin, J. Nguyen, A. Persat, S. Desmarais, M. VanNieuwenhze, K. Huang, J. Zhu, J. Shaevitz, and Z. Gitai, "A periplasmic polymer curves vibrio cholerae and promotes pathogenesis," Cell **168**, 172–185 (2017).
[29]D. A. Bazylinski and R. B. Frankel, "Magnetosome formation in prokaryotes," Nat. Rev. Microbiol. **2**, 217–230 (2004).
[30]S. Nakamura, "Spirochete flagella and motility," Biomolecules **10**, 550 (2020).
[31]I. Nkamba, C. Mulet, G. Dubey, O. Gorgette, A. Couesnon, A. Salles, M. Moya-Nilges, V. Jung, V. Gaboriau-Routhiau, I. Guerrera, T. Shima, Y. Umesaki, G. Nigro, J. Krijnse-Locker, M. Bérard, N. Cerf-Bensussan, P. Sansonetti, and P. Schnupf, "Intracellular offspring released from SFB filaments are flagellated," Nat. Microbiol. **5**, 34–39 (2019).


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[32]M. Raatz, M. Hintsche, M. Bahrs, M. Theves, and C. Beta, "Swimming patterns of a polarly flagellated bacterium in environments of increasing complexity," Eur. Phys. J. Spec. Top. **224**, 1185–1198 (2015).
[33]D. Murat, "Magnetosomes: How do they stay in shape," J. Mol. Microbiol. Biotechnol. **23**, 81–94 (2013).
[34]S. Klumpp and D. Faivre, "Magnetotactic bacteria," Eur. Phys. J. Spec. Top. **225**, 2173–2188 (2016).
[35]V. Nourian and H. Shum, "A numerical method for the locomotion of bi-flagellated bacteria in viscous fluid," Flow **3**, E4 (2023).
[36]C. Rao, J. Kirby, and A. Arkin, "Design and diversity in bacterial chemotaxis: A comparative study in *Escherichia coli* and bacillus subtilis," PLoS Biol. **2**, e49 (2004).
[37]S. Olson, S. Lim, and R. Cortez, "Modeling the dynamics of an elastic rod with intrinsic curvature and twist using a regularized Stokes formulation," J. Comput. Phys. **238**, 169–187 (2013).
[38]W. Lee, Y. Kim, C. Peskin, and S. Lim, "A novel computational approach to simulate microswimmers propelled by bacterial flagella," Phys. Fluids **33**, 111903 (2021).
[39]J. Higdon, "A hydrodynamic analysis of flagellar propulsion," J. Fluid Mech. **90**, 685–711 (1979).
[40]E. Cosserat and F. Cosserat, *Theory of Deformable Bodies* (A. Hermann et Fils, Paris, 1909).
[41]S. S. Antman, *Nonlinear Problems of Elasticity* (Springer-Verlag, New York, 1995).
[42]N. Darnton and H. Berg, "Force-extension measurements on bacterial flagella: Triggering polymorphic transformations," Biophys. J. **92**, 2230–2236 (2007).
[43]S. Fujime, M. Maruyama, and S. Asakura, "Flexural rigidity of bacterial flagella studied by quasielastic scattering of laser light," J. Mol. Biol. **68**, 347–359 (1972).
[44]S. Trachtenberg and I. Hammel, "The rigidity of bacterial flagellar filaments and its relation to filament polymorphism," J. Struct. Biol. **109**, 18–27 (1992).
[45]H. Hoshikawa and R. Kamiya, "Elastic properties of bacterial flagellar filaments. II. Determination of the modulus of rigidity," Biophys. Chem. **22**, 159–166 (1985).
[46]P. Wang, C. Chen, Q. Wang, H. Chen, C. Chen, J. Xu, X. Wang, and T. Song, "Tumor inhibition via magneto-mechanical oscillation by magnetotactic bacteria under a swing MF," J. Controlled Release **351**, 941–953 (2022).
[47]M. A. Constantino, M. Jabbarzadeh, H. C. Fu, Z. Shen, J. G. Fox, F. Haesebrouck, S. K. Linden, and R. Bansil, "Bipolar lophotrichous *Helicobacter Suis* combine extended and wrapped flagella bundles to exhibit multiple modes of motility," Sci. Rep. **8**, 14415 (2018).
[48]P. J. Padgett, M. W. Friedman, and N. R. Krieg, "Straight mutants of *Spirillum volutans* can swim," J. Bacteriol. **153**, 1543–1544 (1983).
[49]M. A. Swan, "Electron microscopic observations of structures associated with the flagella of *Spirillum volutans*," J. Bacteriol. **161**, 1137–1145 (1985).

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